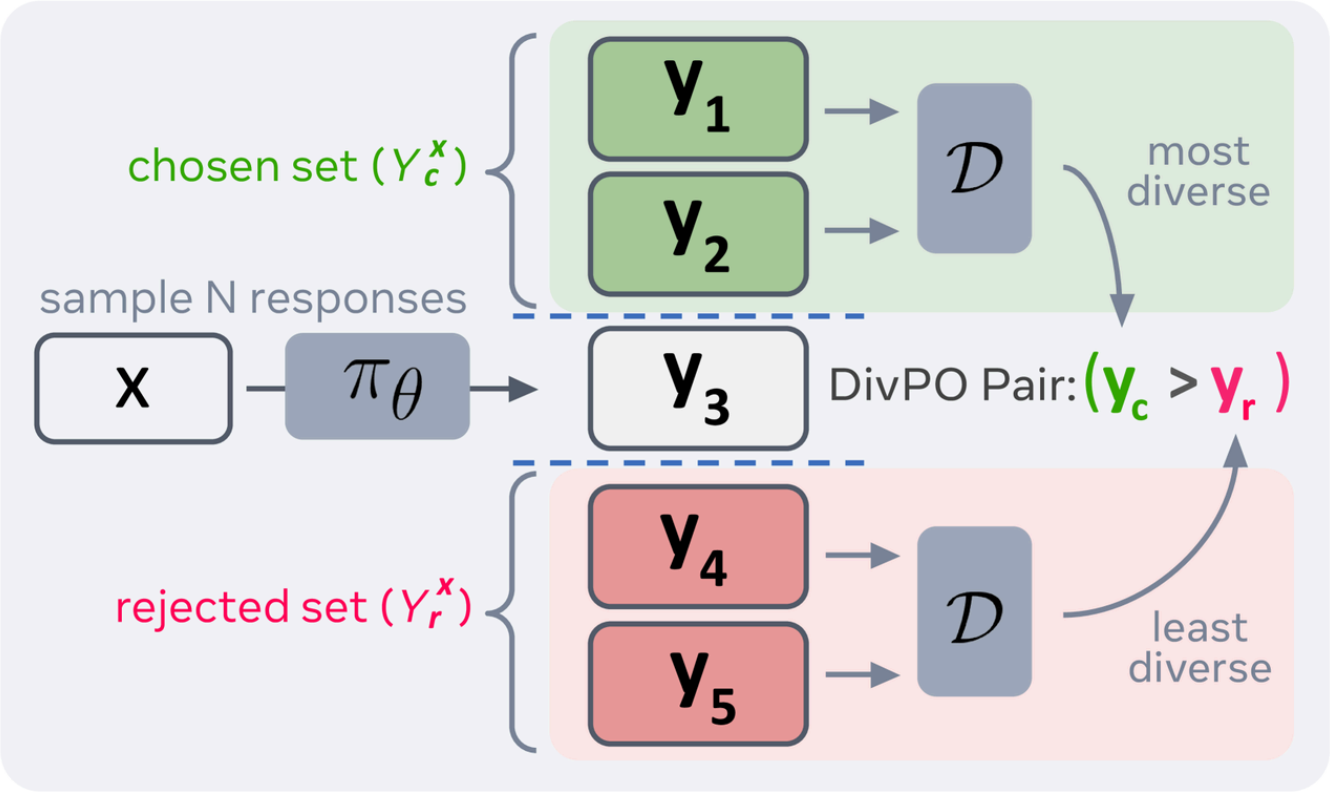
Diverse Preference Optimization

> #Finetuning #DPO



- What problems does this framework solve?
  - After classic RLHF and DPO processes, the LLM will suffer from Alignment Collapse.
  - That is to say, the LLM tends to produce monotonous and lack-of-diversity outputs after the RL/finetuning as it learns that this pattern of answer will cause a low penalty.
- What does this framework do?
  - This framework is a modified version of DPO: in normal DPO, we only consider the responses with the HIGHEST and LOWEST score as the +- pair.
  - Here, we not only consider the score, but also consider the diversity of the responses.
  - Given a prompt x, sample a set of responses $\{y_i\}$.
  - Let the reward model evaluate each response by giving it a score, let the highest score be $s_{\max}$ and the lowest score be $s_{\min}$.
  - Bucketing: We set a threshold score ρ, and let the answers with score $\in [s_{\max} - \rho, s_{\max}]$ be the winner bucket and score $\in [s_{\min}, s_{\min} + \rho]$ be the loser bucket.
  - In the winner bucket, we select the most diverse answer as the + response and the least diverse answer in the loser bucket as the - response. How to measure diversity?
    - Model Probability: a diverse response means the model has a lower possibility to generate that response (rare), so we can use the probability of a response to measure the diversity: $\mathcal{D}(y_i) = -log\ \pi_\theta(y_i|x)$, when this value is large, that answer is rare.
    - Word Frequency: The frequency of vocabulary included in statistical responses within the

current set of responses. Responses containing more high-frequency words exhibit lower diversity.
        - LLM-as-judge.
    - Run DPO.
- Experiment Results
    1. Persona Generation:
        - Requires the model to generate JSON persona cards containing name, city, and occupation.
        - Results: The baseline model (Llama 3.1 Instruct) and DPO tend to repeatedly generate a small number of names (e.g., “Astrid”). DivPO increased attribute diversity by 45.6% while maintaining exceptionally high format accuracy (quality).
    2. Keyword Story Generation:
        - Requires listing 5 keywords suitable for a given title. This is an open-ended creative task.
        - Results: DivPO significantly increased the uniqueness of generated vocabulary (Unique 1-grams) while maintaining similar quality (Win Rate) to DPO. Diversity improved by 74.6% compared to DPO.
    3. General Instruction Following (AlpacaEval 2.0):
        - Trained using the Wildchat dataset.
        - Results: On the general task, DivPO not only increased diversity by 46.2%, but its win rate even surpassed DPO's by 2.4%. This demonstrates that exposure to more diverse “good answers” during training enhances the model's generalization ability and ultimately improves its quality.
- My Thoughts
    - The diversity measurement mechanisms in this framework are not very robust: the rarest answer is not always the best (may contain some hallucination), especially when using the probability method and the word frequency method; the LLM-as-judge method is also highly model-subjective: if the model does not judge well or has some bias, the framework will also collapse (perhaps even worse than the normal DPO).
    - One possible adjustment is using the idea proposed by this paper, as it has factuality checks plus the diversity of the factuality measurements.
- Work Cited
    - Paper
    - Chat History
    - NotebookLM